

Question:

What is the de Broglie wavelength of an electron with a kinetic energy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

To find the de Broglie wavelength of an electron with a kinetic energy of 1 eV , we need to follow these steps :

- 1 . Convert the kinetic energy from eV to Joules : $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$
- 2 . Use the kinetic energy formula : $KE = (1/2)mv^2$, where m is the mass of the electron ($9.11 \times 10^{-31} \text{ kg}$) and v is the velocity of the electron .
- 3 . Rearrange the formula to solve for v : $v = \sqrt{2 KE/m}$
- 4 . Plug in the values : $v = \sqrt{2 \times 1.602 \times 10^{-19} \text{ J} / 9.11 \times 10^{-31} \text{ kg}}$
- 5 . Simplify the expression : $v = \sqrt{2.816 \times 10^{12} \text{ m}^2/\text{s}^2} = 1.685 \times 10^6 \text{ m/s}$
- 6 . Use the de Broglie wavelength formula : $\lambda = h / p$, where h is the Planck constant ($6.626 \times 10^{-34} \text{ J s}$) and p is the momentum of the electron (mv).
- 7 . Plug in the values : $\lambda = 6.626 \times 10^{-34} \text{ J s} / (9.11 \times 10^{-31} \text{ kg} \times 1.685 \times 10^6 \text{ m/s})$
- 8 . Simplify the expression : $\lambda = 4.10 \times 10^{-9} \text{ m}$

Now , let's convert the wavelength from meters to centimeters : $\lambda = 4.10 \times 10^{-9} \text{ m} \times (100 \text{ cm} / 1 \text{ m}) = 4.10 \times 10^{-7} \text{ cm}$

The closest answer choice is A. $2.48 \times 10^{-8} \text{ cm}$, but the calculated value is $4.10 \times 10^{-7} \text{ cm}$, which is not among the options .